

Lesson 4: Sample Integration & Comparative Analysis

Spatial Transcriptomics Course

Learning Objectives

- Integrate multiple spatial transcriptomics samples
- Remove technical batch effects while preserving biology
- Identify differentially expressed genes between conditions
- Perform gene set enrichment analysis (GSEA) with KEGG
- Compare cell type proportions across conditions
- Visualize treatment effects in spatial context
- Align tissue slices to a shared coordinate frame with STalign
- Apply statistical testing with appropriate corrections
- Interpret results in biological context

Part 1: Introduction to Sample Integration

Why integrate? Understanding batch effects

Why Integrate Multiple Samples?

- Biological Questions:

- Treatment vs control effects
- Disease vs healthy tissue
- Developmental time courses
- Spatial replicates
- Multi-region studies

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- Benefits of Integration:

- Increased statistical power
- More robust conclusions
- Control for variability
- Enable meta-analysis

- Technical Challenges:

- Batch effects dominate signal
- Sequencing depth differences
- Processing date variations
- Operator-specific effects
- Platform differences

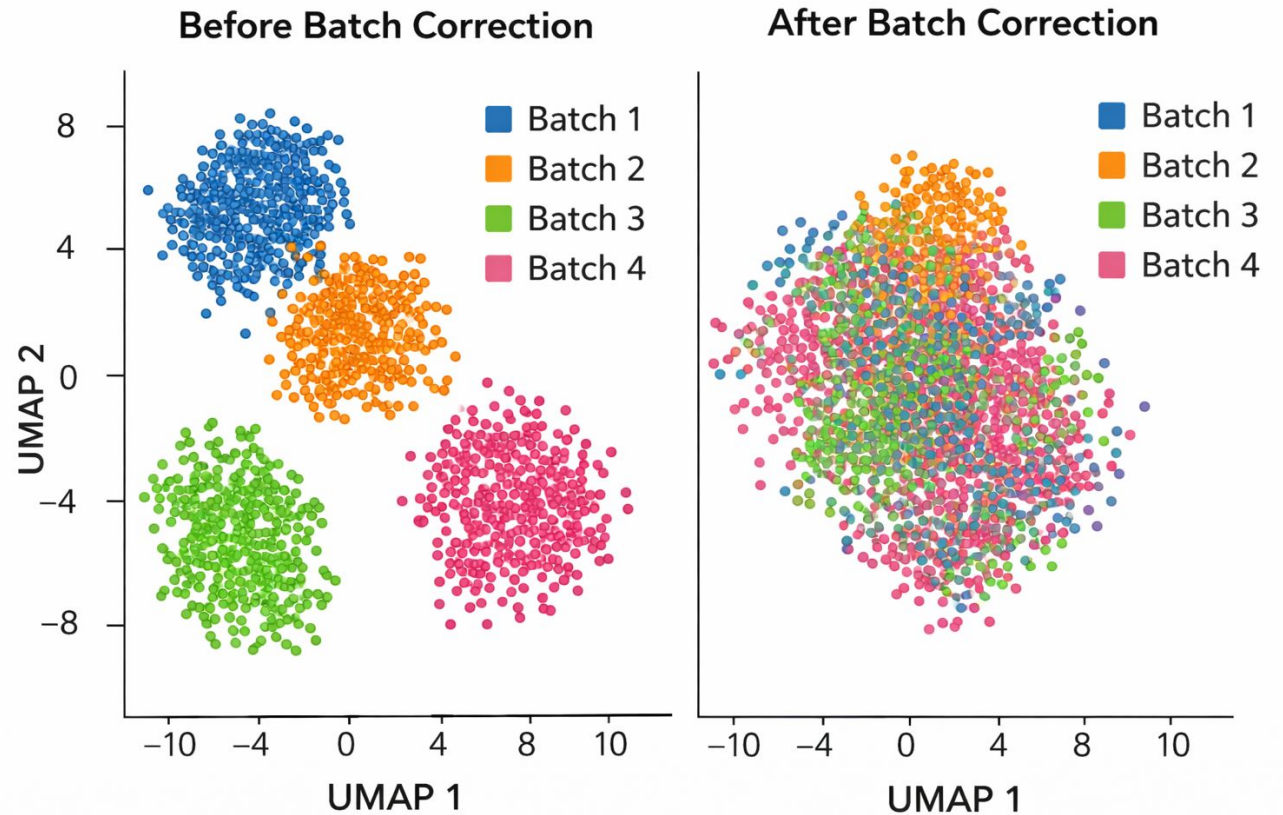
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- Solution: Integration

- Remove technical variation
- Preserve biological signal
- Enable fair comparison
- Align samples in embedding space

Understanding Batch Effects

- What are batch effects?
 - Technical variations unrelated to biology
 - Can completely dominate biological signal
 - Cause spurious differential expression
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- Common Sources:
 - Different processing dates or operators
 - Sequencing depth variations across runs
 - Library preparation batch differences
 - Reagent lot variations
 - Platform or protocol changes
- Detection:
 - Samples cluster by batch, not biology
 - Known markers show unexpected patterns
 - Strong PC correlates with batch ID



Part 2: Harmony Integration Algorithm

Fast, effective batch correction for spatial data

Harmony Algorithm Overview

- Key Features:
 - Works directly on PCA embeddings
 - Iterative soft clustering and correction
 - Handles multiple batches simultaneously
 - Computationally efficient and scalable
- Algorithm Steps:
 - 1. Cluster cells across all batches
 - 2. For each cluster, compute batch-specific centroids
 - 3. Move cells toward global centroid (soft assignment)
 - 4. Repeat until convergence
- Advantages:
 - Preserves cell type structure
 - Minimal parameter tuning needed
 - Much faster than Mutual Nearest Neighbors (MNN) or Seurat
 - Works well with spatial data

Sample Integration Workflow

- Step 1: Preprocess individual samples
 - QC filtering, normalization, log-transform
 - Same pipeline for all samples
- Step 2: Concatenate samples
 - Combine into single AnnData object
 - Track sample/batch identity in metadata
- Step 3: Feature selection
 - Identify highly variable genes across all samples
 - Typically 3000-5000 genes
- Step 4: Compute PCA
 - Dimensionality reduction on combined data
 - Usually 50 principal components
- Step 5: Run Harmony
 - Correct PC coordinates using batch labels
 - Returns batch-corrected PCs
- Step 6: Downstream analysis
 - Use corrected PCs for neighbors, UMAP, clustering, etc.

Evaluating Integration Quality

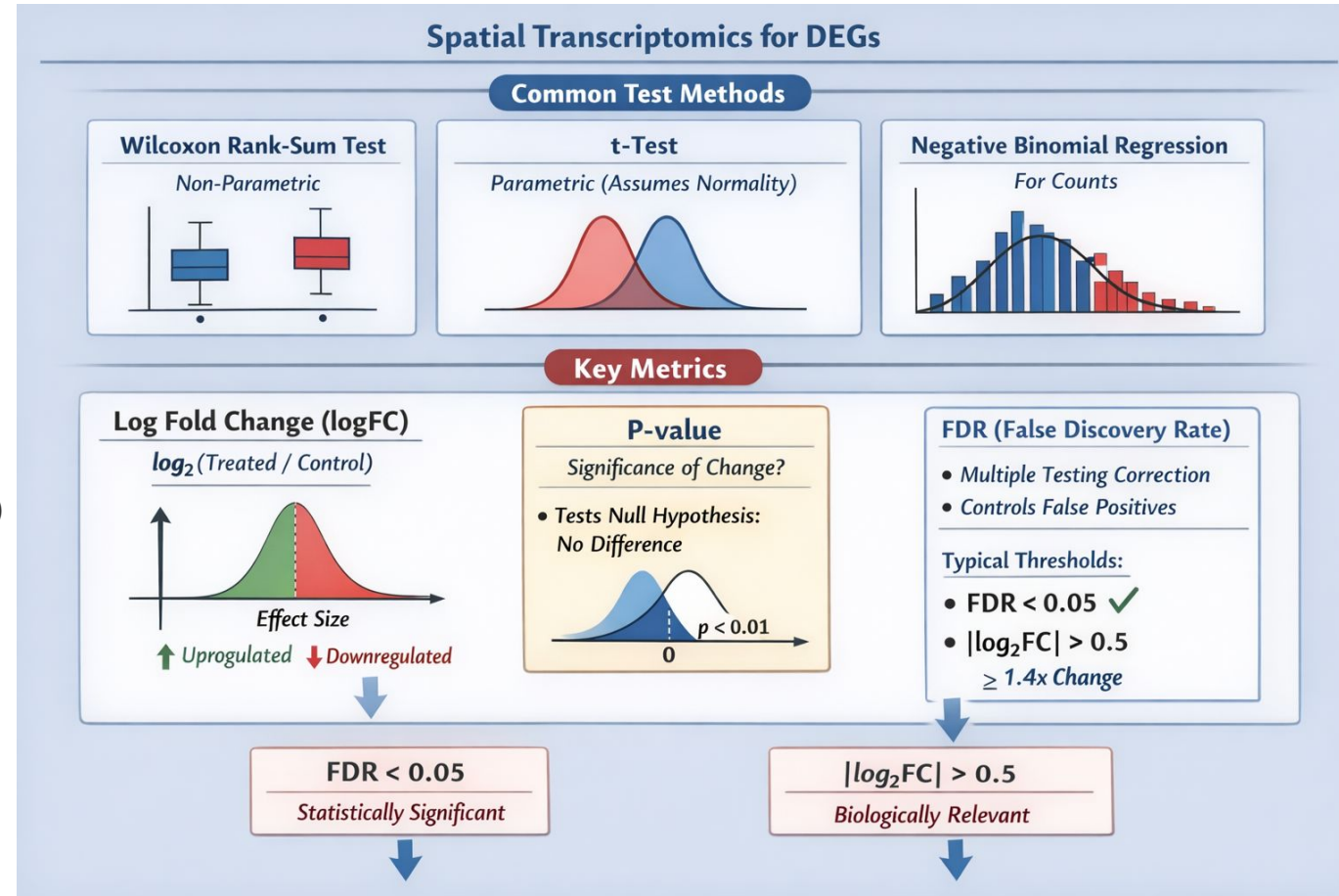
- Visual Inspection:
 - UMAP colored by batch (before/after)
 - UMAP colored by condition
 - UMAP colored by cell type markers
- Before Integration:
 - Samples separate by batch
 - Technical variation dominates
 - Cannot fairly compare conditions
- After Integration:
 - Batches mix within cell types
 - Biological structure preserved
 - Fair comparison possible
- Quantitative Metrics:
 - Batch mixing score (kBET)
 - Silhouette coefficient by cell type
 - Average Silhouette Width (ASW)
 - Local Inverse Simpson's Index (LISI)
- Warning Signs:
 - Over-correction
 - Loss of biological signal (metrics)
- Best Practice:
 - Always check known marker genes
 - Validate rare cell types survive
 - Compare multiple integration methods

Part 3: Differential Expression Analysis

Identifying genes that change between conditions

Statistical Testing for DEGs

- Common Test Methods:
 - Wilcoxon rank-sum test (non-parametric)
 - t-test (parametric, assumes normality)
 - Negative binomial regression (for counts)
- Log Fold Change (logFC):
 - Effect size - how much does expression change?
 - $\log_2(\text{Treated}/\text{Control})$
 - Positive = upregulated, Negative = downregulated
 - $|\log_2\text{FC}| > 0.5$ ($\approx 1.4\text{x}$ change, biological relevance)
- P-value:
 - Statistical significance - is change real?
 - Tests null hypothesis: no difference
- FDR (False Discovery Rate):
 - Multiple testing correction
 - Controls expected proportion of false positives
 - Typical Threshold: $\text{FDR} < 0.05$ (statistical significance)

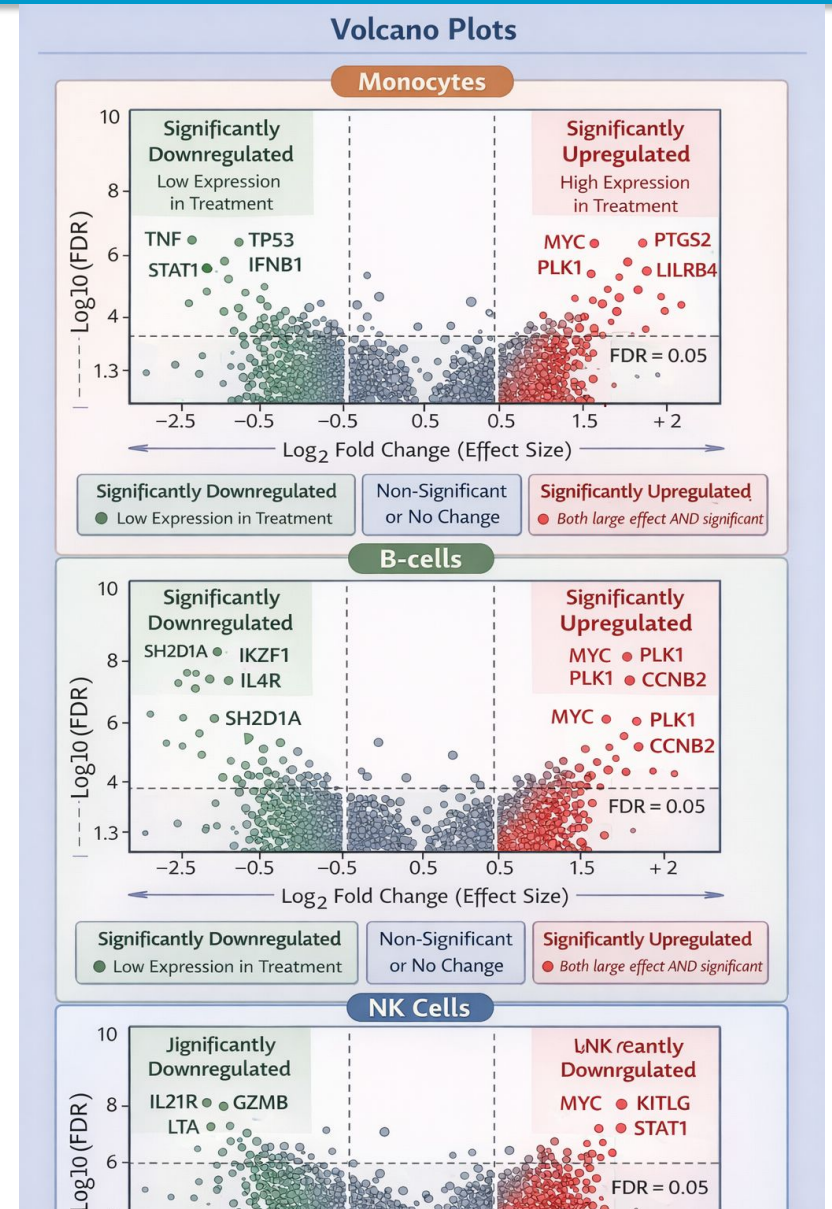


Which test to use?

Feature	t-test	Wilcoxon Rank-Sum
Parametric	Yes	No 
Requires normal distribution	Yes	No
Robust to outliers	No	Yes
Handles zero inflation	Poorly	Better
Works with small samples	Less reliable	Good
Metric used	Mean difference	Rank ordering

Volcano Plots: Visualizing Differential Expression

- Plot Structure:
 - X-axis: Log2 fold change (effect size)
 - Y-axis: $-\log_{10}(\text{FDR})$ (significance)
 - Each point is one gene
- Top-right corner: Significantly upregulated
 - High expression in treatment - Both large effect AND significant
- Top-left corner: Significantly downregulated
 - Low expression in treatment - Both large effect AND significant
- Center-bottom: Non-significant or no change
- Best Practices:
 - Color-code by significance status
 - Label top genes for readability
 - Include threshold lines for clarity
 - Volcano plot per cell type



Part 4: Gene Set Enrichment Analysis

From genes to pathways - biological interpretation

Gene Set Enrichment Analysis (GSEA)

- Core Philosophy:

- Analyze pathways, not individual genes
- Coordinated modest changes matter
- Group statistics increase power

- GSEA Algorithm:

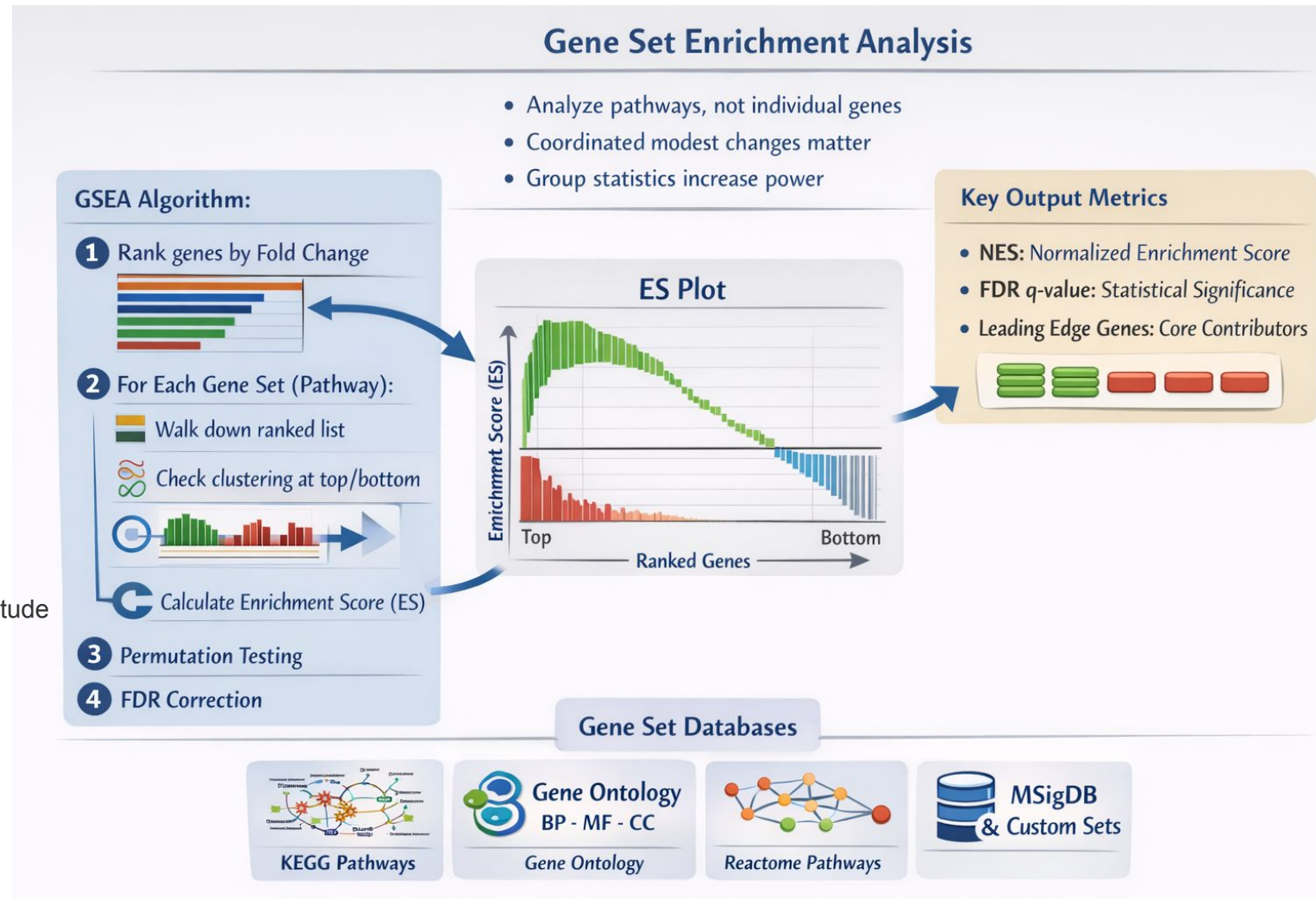
1. Rank all genes by fold change (or other metric)
2. For each gene set (pathway):
 - - Walk down ranked list
 - - Check if gene set members cluster at top/bottom
 - - Calculate enrichment score (ES)
3. Permutation testing for significance
4. FDR correction across all gene sets

- Key Output Metrics:

- NES (Normalized Enrichment Score): direction and magnitude
- FDR q-value: statistical significance
- Leading edge genes: core contributors

- Gene Set Databases:

- KEGG pathways (metabolism, signaling)
- Gene Ontology (BP, MF, CC)
- Reactome, MSigDB, custom sets



KEGG Pathways for Biological Interpretation

- What is KEGG?
 - Kyoto Encyclopedia of Genes and Genomes
 - Curated biological pathway database
 - 500+ pathways covering all biology
 - Expert-curated from literature
- Pathway Categories:
 - Metabolism
 - Glucose, lipids, amino acids
 - Genetic Information
 - Transcription, translation
 - Environmental Information
 - Signal transduction
 - Cellular Processes
 - Cell cycle, apoptosis
- Using KEGG with GSEA:
 - Test enrichment for each pathway
 - Identify activated/suppressed processes
 - Generate testable hypotheses
- Example Interpretation: Finding: 20 upregulated, 30 downregulated genes
 - Hard to interpret individually
- GSEA reveals:
 - 'Oxidative phosphorylation' ↓ , 'Glycolysis' ↑
- Biological Story:
 - Warburg effect - metabolic switch
 - Cells shifting from mitochondrial to glycolytic metabolism
 - Coherent, testable hypothesis
- Power of Pathway Analysis:
 - Works even when individual genes not significant
 - 50 genes changing 1.3x each → significant pathway

Interpreting GSEA Results

- NES (Normalized Enrichment Score):
 - Positive: pathway upregulated in condition
 - Negative: pathway downregulated
 - $|\text{NES}| > 1.5$ typically meaningful
- FDR q-value:
 - < 0.25 : significant (more lenient than DEG)
 - < 0.05 : highly significant
- Interpretation Strategy:
 - Look for coherent biological stories
 - Multiple related pathways changing together = strong signal
 - Context matters - same result can mean different things
 - Always check leading edge genes
- Example - Cancer Treatment:
 - Cell cycle pathways $\downarrow \rightarrow$ Good (blocking proliferation)
 - Immune pathways $\uparrow \rightarrow$ Good (immune activation)
 - DNA repair $\downarrow \rightarrow$ Treatment working

Part 5: Cell Type Proportion Analysis

Does treatment change tissue composition?

Analyzing Cell Type Proportions

- Analysis Steps:

- 1. Cluster integrated data into cell types
- 2. Count cells per type per condition
- 3. Calculate proportions (% of total)
- 4. Statistical testing
- 5. Visualize changes spatially

- Immune cell infiltration:

- Treatment recruits T cells into tumor

- Cell death/proliferation:

- Neurodegenerative disease → neuron loss

- Differentiation state changes:

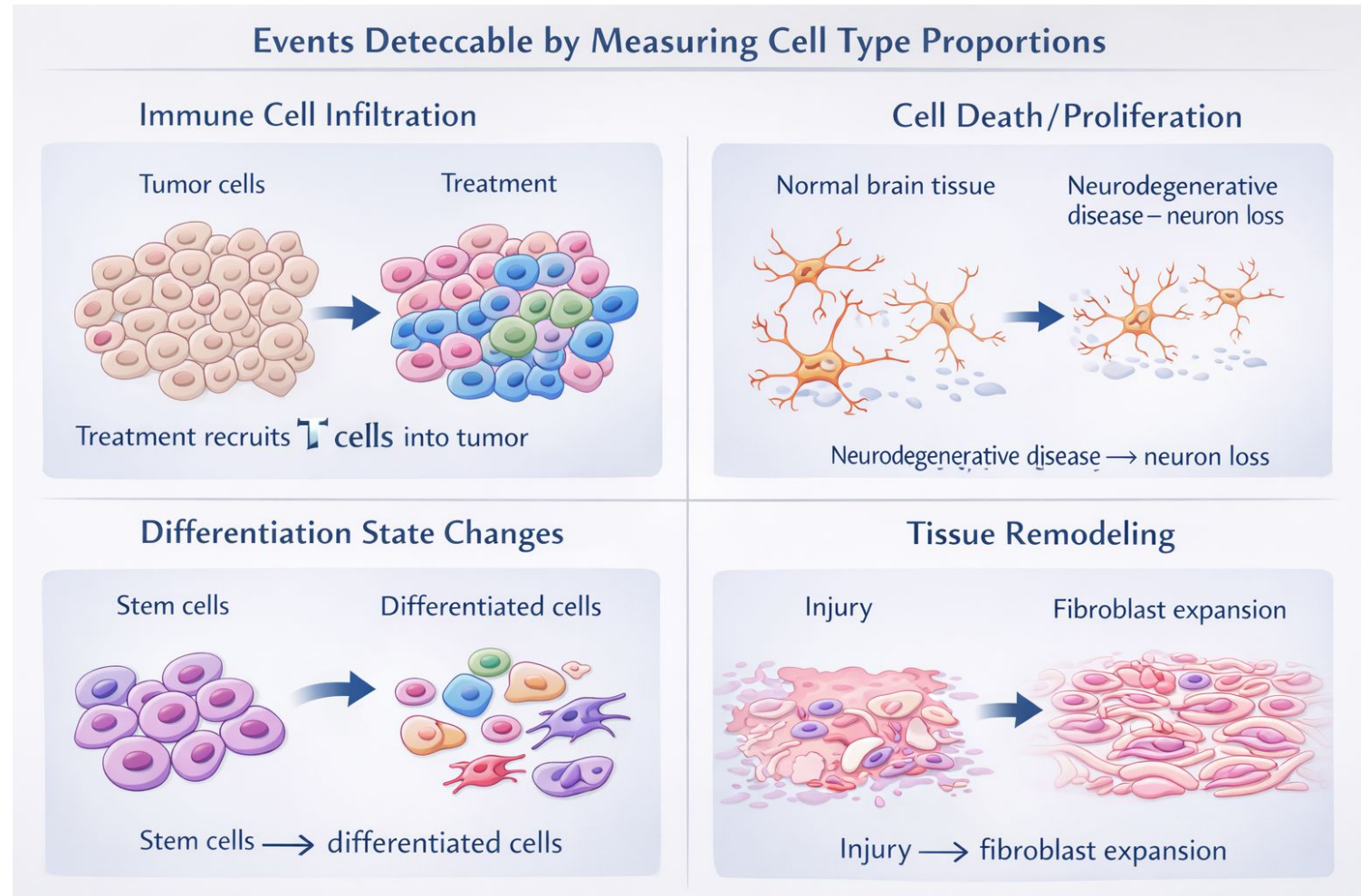
- Stem cells → differentiated cells

- Tissue remodeling:

- Injury → fibroblast expansion

- Important Considerations:

- Is change due to cell gain or other cell loss?
- Check proliferation and apoptosis markers
- Spatial patterns provide mechanism clues



Statistical Testing for Proportions

- Chi-square Test:
 - Tests overall distribution
 - Null: proportions same across conditions
 - Returns global p-value
- If significant:
 - At least one cell type differs
 - Doesn't tell you which one
- Requirements:
 - Sufficient cell counts
 - Independent observations
 - At least 5 cells per category
- Interpretation:
 - $p < 0.05$: distribution differs globally
 - Follow up with Fisher's tests
- Fisher's Exact Test:
 - Tests each cell type individually
 - 2×2 table: this type vs all others
 - Exact p-value (no approximation)
- Advantages:
 - Works with small counts
 - Identifies specific cell types
 - Provides odds ratios
- Multiple Testing Correction:
 - Test many cell types → many p-values
 - Use FDR (Benjamini-Hochberg)
 - Controls false discovery rate
- Odds Ratio Interpretation:
 - OR = 1: no enrichment
 - OR = 2: 2× more likely in treated
 - OR = 0.5: 2× less likely in treated

Part 6: Spatial Domain Alignment

Registering tissue slices with STalign

STalign Under the Hood: Problem Formulation

- The Alignment Problem:

- Two slices: source $S(x,y)$ and target $T(x,y)$
- Coordinates are NOT comparable - different frames
- Goal: find warp ϕ so that $\phi(S) \approx T$ in physical space

- Tissue Representation:

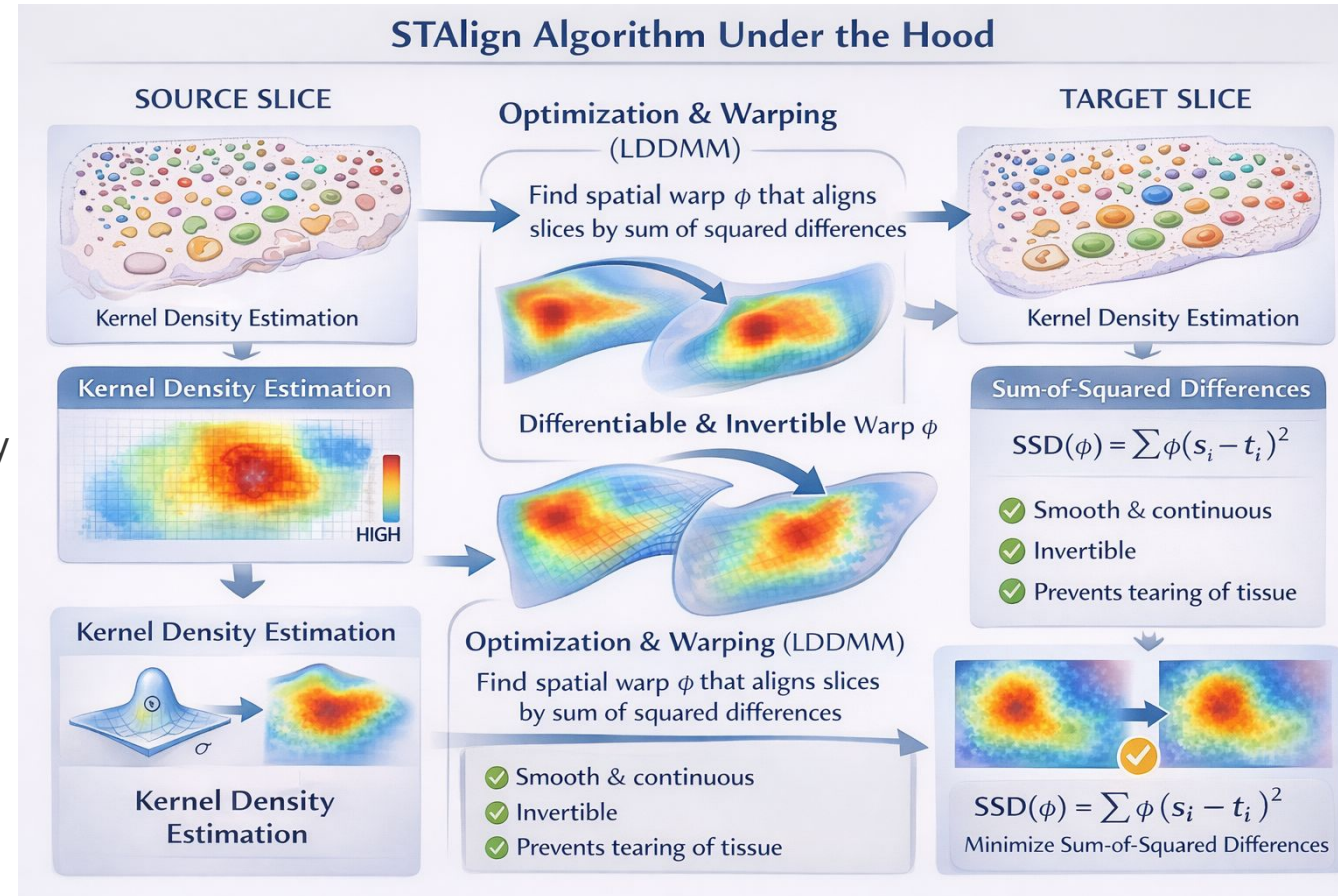
- Each slice is converted to a 2D spatial density field
- Kernel density estimation over spot positions
- Optional: weight by gene expression or cell-type probability
- Two continuous images $I_S(x,y)$ and $I_T(x,y)$ to align

- Similarity Metric:

- Sum-of-squared differences between density images
- $\|I_S(\phi(x,y)) - I_T(x,y)\|^2$ minimised over ϕ
- Captures spatial overlap of tissue structure

- Framework: LDDMM

- Large Deformation Diffeomorphic Metric Mapping
- Allows large, smooth, invertible deformations
- Standard tool in medical image registration



STalign Under the Hood: Diffeomorphic Optimisation

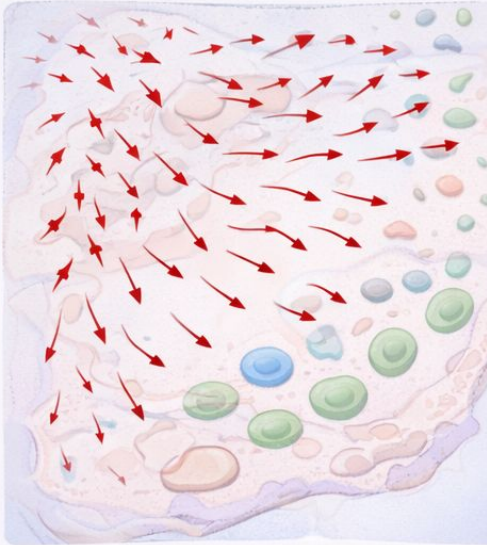
- Velocity Field Parameterisation:
 - Warp ϕ is NOT specified directly
 - Instead: define time-varying velocity field $v(x, t)$
 - Integrate ODE $dx/dt = v(x, t)$ from $t=0$ to $t=1$
 - ϕ = flow of $v \rightarrow$ always diffeomorphic
- Energy Functional:
 - $E(v) = \text{Data term} + \lambda \cdot \text{Regularisation term}$
 - Data: $\|I_S(\phi) - I_T\|^2$ (alignment quality)
 - Reg: $\|Lv\|^2$ (smoothness of deformation)
 - L = differential operator (Laplacian-based)
 - λ controls rigidity vs. flexibility trade-off
- Optimisation:
 - Gradient descent on velocity field parameters
 - Gradient computed via adjoint method (backprop)
 - Scales to millions of spots via GPU acceleration
- Output & What to Do With It:
 - Transformed coordinates (x', y') per spot
 - Store in `adata.obsm['spatial_aligned']`
 - Both slices now live in same coordinate frame
- Downstream Analyses Enabled:
 - Cell-type domain correspondence across sections
 - Region-level DEG analysis (matched spatial zones)
 - Morphological overlay plots
 - Inter-slice cell-communication inference

Spatial Visualization of Treatment Effects

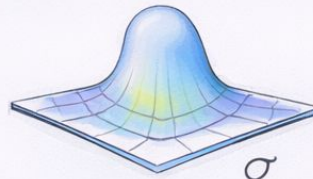
- Why Spatial Matters:
 - See WHERE changes occur in tissue
 - Identify regional vs global effects
 - Understand tissue architecture
 - Generate mechanism hypotheses
- Side-by-side comparison:
 - Control vs Treated, same gene/cell type
 - Use identical color scales
- Gene expression maps:
 - Show DEGs in spatial context
- Cell type distributions:
 - Overlay cluster assignments on tissue
- Statistical overlays:
 - Color by fold change or p-value
- Spatial Patterns Reveal Mechanism:
 - Boundary effects → local processes at interfaces
 - Regional specificity → zone-specific responses
 - Gradients → diffusible signals
 - Hotspots → local production or aggregation

Interpreting Spatial Warp Outputs from STAlign

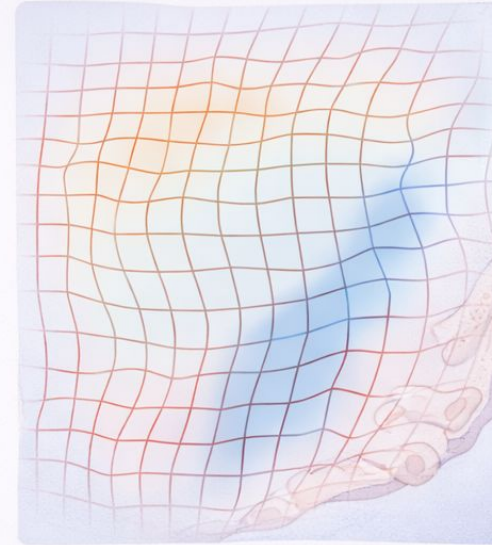
Displacement Field



Multiple segmentant φ

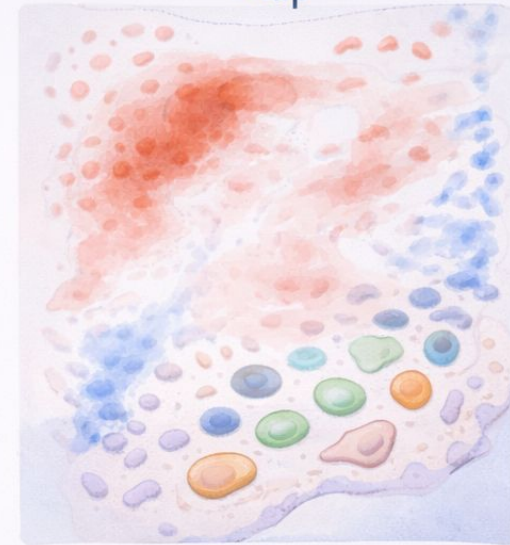


Deformation Grid



Warped grid illustrating φ

Expansion or Compression Map



Jacobian determinant, $J_{\varphi}(x)$



Expansion Stable Compression

Best Practices & Common Pitfalls

- Always Do:

- ✓ Visually inspect integration quality
- ✓ Use biological replicates if possible
- ✓ Set appropriate statistical thresholds
- ✓ Apply multiple testing correction
- ✓ Validate top DEGs independently
- ✓ Check for known marker genes
- ✓ Visualize results spatially
- ✓ Interpret in biological context
- ✓ Report all methods and parameters
- ✓ Compare multiple integration methods

- Never Do:

- ✗ Skip visual quality control
- ✗ Compare without integration
- ✗ Ignore batch effects
- ✗ Report uncorrected p-values
- ✗ Cherry-pick favorable pathways
- ✗ Over-interpret borderline results
- ✗ Trust outlier genes without validation
- ✗ Forget biological plausibility
- ✗ Lower thresholds to get 'significance'
- ✗ Skip replication if possible

Hands-on Exercise: Complete Workflow

- Dataset:
 - Control: `adult_mouse_brain_ST4k.h5ad`
 - Treated: `adult_mouse_brain_ST4k_treated.h5ad` (Mock treatment effects simulated)
- Complete Analysis Tasks:
 - 1. Load and QC both samples
 - 2. Integrate with Harmony
 - 3. Identify differentially expressed genes
 - 4. Create volcano plot
 - 5. Run GSEA with KEGG pathways
 - 6. Analyze cell type proportion changes
 - 7. Visualize top DEGs spatially
 - 8. Align slices with STalign and compare spatial domains
 - 9. Create publication-quality figures
- Tools & Resources:
 - Jupyter notebook: `sample_integration.ipynb`
 - Python packages: `scanpy`, `harmonypy`, `gseapy`, `stalign`
 - Step-by-step guidance provided

Additional Resources

- Software & Documentation:

- Scanpy: <https://scanpy.readthedocs.io>
- Harmony: <https://github.com/Slowkow/harmony>
- GSEAPy: <https://gseapy.readthedocs.io>

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- Pathway Databases:

- KEGG: <https://www.genome.jp/kegg/>
- Gene Ontology: <http://geneontology.org>
- MSigDB: <https://www.gsea-msigdb.org>
- Reactome: <https://reactome.org>

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- Key Publications:

- Harmony: Korsunsky et al. Nature Methods 2019
- GSEA: Subramanian et al. PNAS 2005
- Spatial transcriptomics review: Moses & Pachter Nat Rev Genet 2022

Course summary questions

Which are two major challenges in spatial transcriptomics?

What is BANKSY and why is it used instead of standard clustering?

What is the traditional workflow for cell type annotation?

Which are advantages of reference-based cell type annotation over the traditional workflow?

What are the main reference-based annotation methods covered, and what does each do?

What is Hotspot?

What is spatial domain detection and which tools are used?

How does spatial ligand-receptor (L-R) analysis work in Squidpy/CellPhoneDB?

What are batch effects and why must they be corrected before comparing samples?

What is STalign and what problem does it solve?

Questions?

Thank you!